Progress test 3

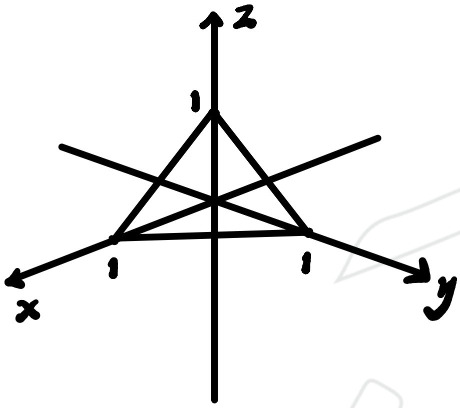
1. Surface integral and integral theorem

   (a) Let $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ are vertices of a triangle S. If a vector field $\mathbf{F}$ is defined by $\mathbf{F}(x,y,z) = (1, e^z \cos x \sin y, -e^z \cos x \sin y)$, find $\iint_S \mathbf{F} \cdot d\mathbf{S}$. [30%]

   (b) Sketch a surface S given by $z = -x^2 - 4y^2 + 4$ for $z \geq 0$. If a vector field $\mathbf{G}(x,y,z) = xe^{xy}\mathbf{i} + 4ye^{xy}\mathbf{j} + z^3\mathbf{k}$, find $\iint_S (\nabla \times \mathbf{G}) \cdot d\mathbf{S}$. [30%]

   (c) Let a region B in $\mathbb{R}^3$ be a circular cylinder bounded by $(x-5)^2 + (y-2)^2 = 16$, $z = 4$, and $z = 8$. Sketch the region B. If a vector field $\mathbf{H}$ is given by $\mathbf{H}(x,y,z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, find $\iint_{\partial B} \mathbf{H} \cdot d\mathbf{S}$ where ∂B is the boundary surface of B. [40%]

(a)



$x+y+z=1$

$\therefore f = x+y+z = 1$

$\therefore n = \dfrac{\nabla f}{\|\nabla f\|} = \dfrac{(1,1,1)}{\sqrt{1^2+1^2+1^2}}$

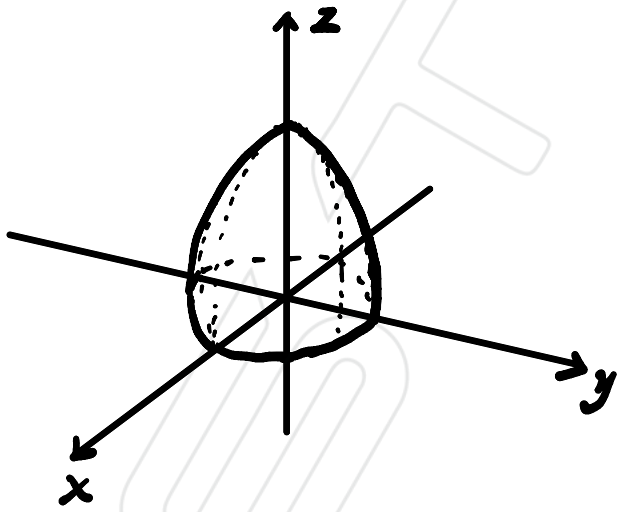
$= \left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right)$

$\therefore \iint_S F \cdot dS = \iint_S F \cdot n \, dA$

$= \iint_S \left(1, e^{\frac{\sqrt{3}}{3}} \cos(\frac{\sqrt{3}}{3}) \sin(\frac{\sqrt{3}}{3}), -e^{\frac{\sqrt{3}}{3}} \cos(\frac{\sqrt{3}}{3}) \sin(\frac{\sqrt{3}}{3})\right)\left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right) dA$

$= \iint_S \frac{\sqrt{3}}{3} dA = \frac{\sqrt{3}}{3} A = \frac{\sqrt{3}}{3}\left(\frac{1}{2}\|a\|\|b\| \sin\theta\right) = \frac{\sqrt{3}}{6}\|a \times b\|$

$= \frac{\sqrt{3}}{6}\|(-1,0,1) \times (0,-1,1)\| = \frac{\sqrt{3}}{6}\|(1,-1,1)\| = \frac{\sqrt{3}}{6} \cdot \sqrt{3} = \frac{1}{2}$



(b) When $y=0$, $z^2=-x^2+4$

When $x=0$, $z^2=-4y^2+4$

When $z=0$, $x^2+4y^2=4$

$\therefore \iint_S (\nabla\times G)\cdot ds$

$= \int_{\partial S} G\cdot ds$

At boundary condition, $z=0$ $\therefore x^2+4y^2=4=(r_1\cos\theta)^2+4(r_2\sin\theta)^2$

$\therefore \partial S=(2\cos\theta, \sin\theta, 0)$

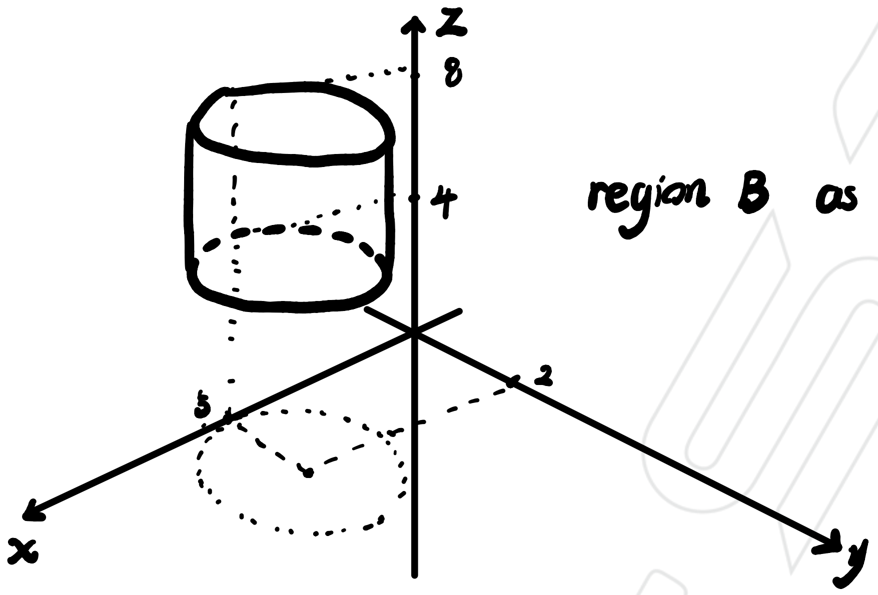
$\therefore \iint_S (\nabla\times G)\cdot ds = \int_{\partial S} G\cdot \frac{d(\partial S)}{d\theta}\, d\theta$

$= \int_0^{2\pi} (xe^{xy}, 4ye^{xy}, z^3)\cdot(-2\sin\theta, \cos\theta, 0)\, d\theta$

$= \int_0^{2\pi} -4\cos\theta\sin\theta e^{xy} + 4\cos\theta\sin\theta e^{xy} + 0 \;d\theta$

$= \int_0^{2\pi} 0 \; d\theta = 0$

(C)



region B as shown

$$\iint_{\delta B} H \cdot ds = \iiint_B \nabla \cdot H \cdot dV$$

for $z = 4$, $(x-5)^2 + (y-2)^2 = 16 = r^2$ $\therefore r = 4$

$\therefore \delta B = (4\cos\theta + 5, 4\sin\theta + 2, 4)$

for $z = 8$, $\delta B = (4\cos\theta + 5, 4\sin\theta + 2, 4)$

$$\therefore \iiint_B \nabla \cdot H \cdot dV = \iiint_B (1+1+1) dV = 3 \iiint_B dV$$

$$= 3 \cdot V = 3 \cdot (\pi r^2) \cdot h$$

$$= 192\pi$$